

Assignment 8: Time Series Analysis

Mickey Zhang

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
#1
```

```
library(here); library(tidyverse); library(zoo); library(ggplot2);
```

```
## here() starts at /home/guest/R/EDE_Fall2025/EDE_Fall2025
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr   1.5.1
```

```
## v ggplot2    4.0.0      v tibble    3.2.1
```

```
## v lubridate  1.9.3      v tidyr     1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
##
```

```
## Attaching package: 'zoo'
```

```
##
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(lubridate);library(trend)
here()
```

```
## [1] "/home/guest/R/EDE_Fall2025/EDE_Fall2025"
```

```
# setting my ggplot theme
mytheme = theme_bw()+
  theme(plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(hjust = 0.5),
        legend.position = "right"
  )
set_theme(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#2
folder = here('Data','Raw','Ozone_TimeSeries')

O3_raw = list.files(path = folder, pattern = "\\*.csv$", full.names = TRUE) %>%
  map_dfr(read_csv, show_col_types = FALSE)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame **GaringerOzone**.

```
# 3
O3_raw$Date = as.Date(O3_raw$Date, format = "%m/%d/%Y")
# 4
O3 = O3_raw %>% select("Date",
                      "Daily Max 8-hour Ozone Concentration",
                      "DAILY_AQI_VALUE")
```

```
# 5
Days = as.data.frame(seq(from = as.Date("2010-01-01"),
                          to   = as.Date("2019-12-31"),
                          by   = "day"))
names(Days) = "Date"
# 6
O3 = left_join(Days,O3, by = "Date")
```

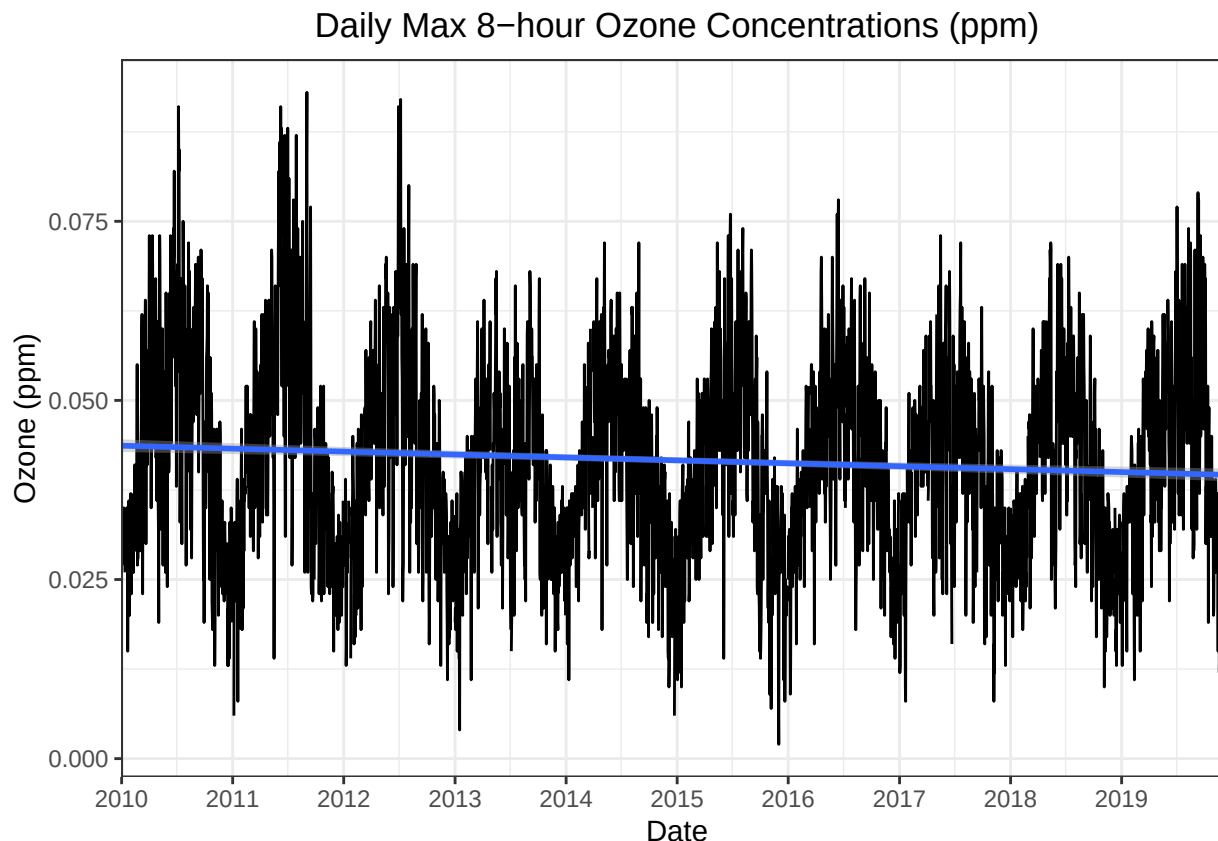
Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
ggplot(O3, aes(x = Date, y=`Daily Max 8-hour Ozone Concentration`))+
  geom_line() +
  geom_smooth(method = "lm")+
  labs(title = "Daily Max 8-hour Ozone Concentrations (ppm)",
       x = "Date", y = "Ozone (ppm)")+
  scale_x_date(
    limits = c(as.Date("2010-01-01"),as.Date("2019-12-31")),
    expand = c(0,0),
    date_labels = "%Y",
    date_breaks = "1 year",
    date_minor_breaks = "6 month"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: The slightly downward-sloping line indicates there's a decreasing trend of ozone concentration overtime.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
O3$`Daily Max 8-hour Ozone Concentration` =
  na.approx(O3$`Daily Max 8-hour Ozone Concentration`, x=O3$Date, na.rm = FALSE)
```

Answer: Piecewise constant fills missing days with the last known value, which is unrealistic for ozone, since concentrations vary continuously with weather and chemistry. Spline interpolation fits data using quadratic function, which can introduce artificial oscillations or negative value.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

#9

```
GaringerOzone.monthly = O3 %>%  
  mutate(Year = year(Date), Month = month(Date)) %>%  
  group_by(Year, Month) %>%  
  summarise(meanO3 = mean(`Daily Max 8-hour Ozone Concentration`)) %>%  
  mutate(Date = as.Date(paste(Year, Month, "01", sep = "-")))
```

'summarise()' has grouped output by 'Year'. You can override using the
'.groups' argument.

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

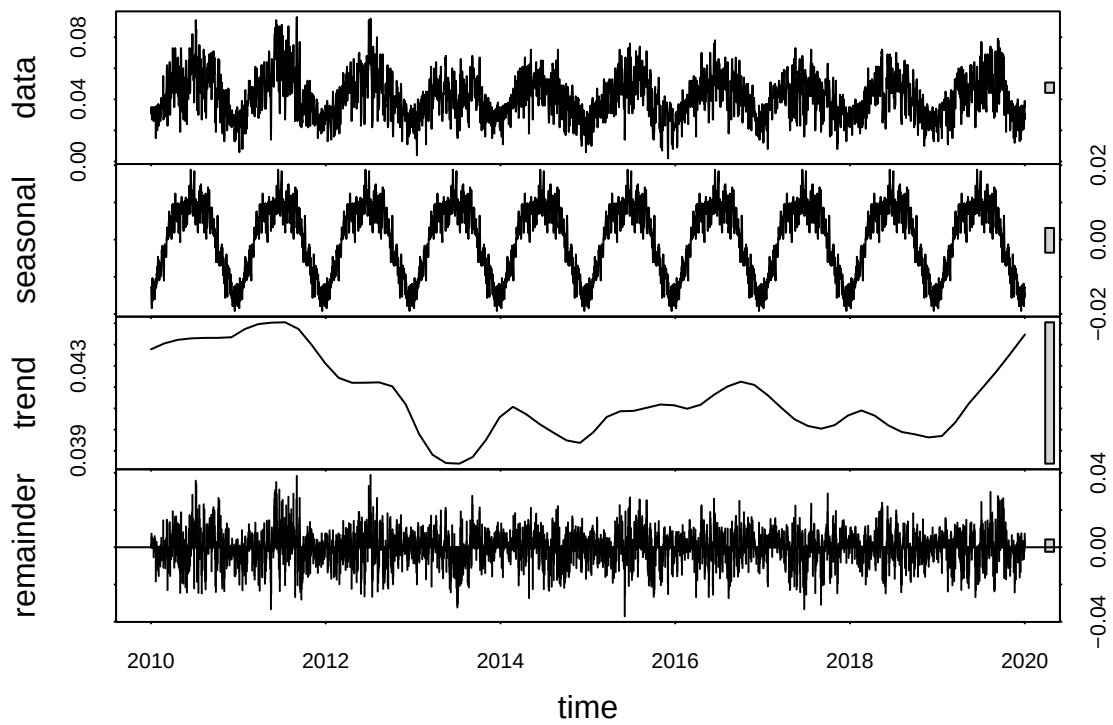
#10

```
GaringerOzone.daily.ts = ts(O3$`Daily Max 8-hour Ozone Concentration`,  
  start = c(2010,1), frequency = 365 )  
GaringerOzone.monthly.ts = ts(GaringerOzone.monthly$meanO3,  
  start = c(2010,1), frequency = 12 )
```

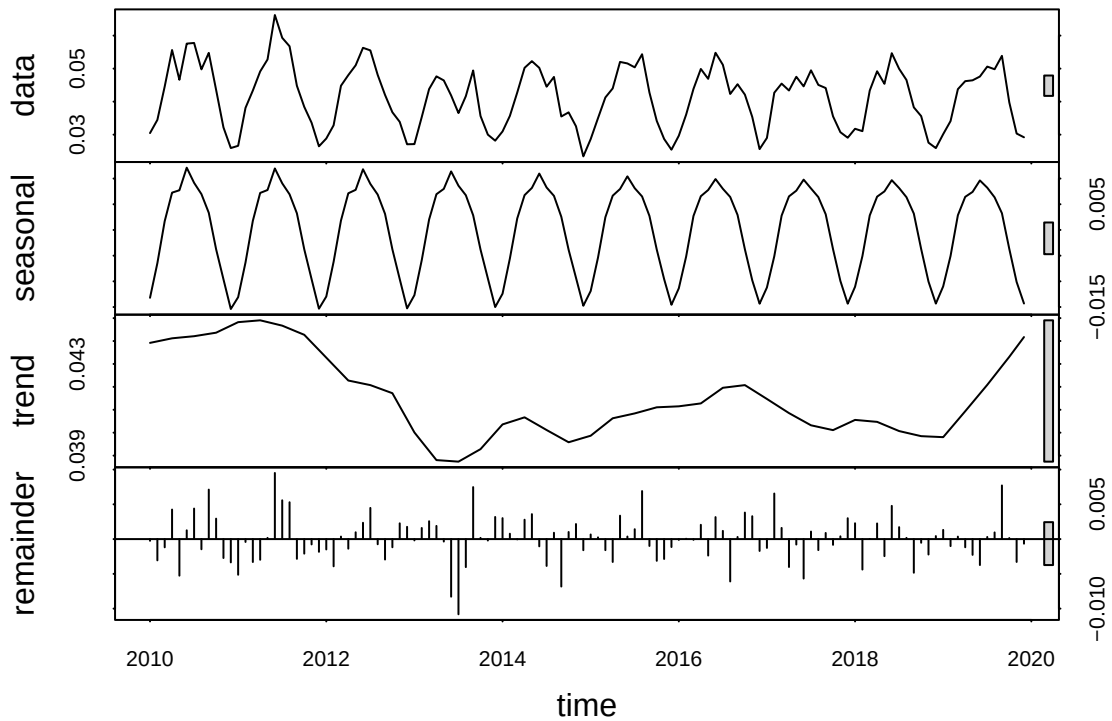
11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

#11

```
daily_decomp = stl(GaringerOzone.daily.ts , s.window = 365)  
plot(daily_decomp)
```



```
monthly_decomp = stl(GaringerOzone.monthly.ts , s.window = 12)
plot(monthly_decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

#12

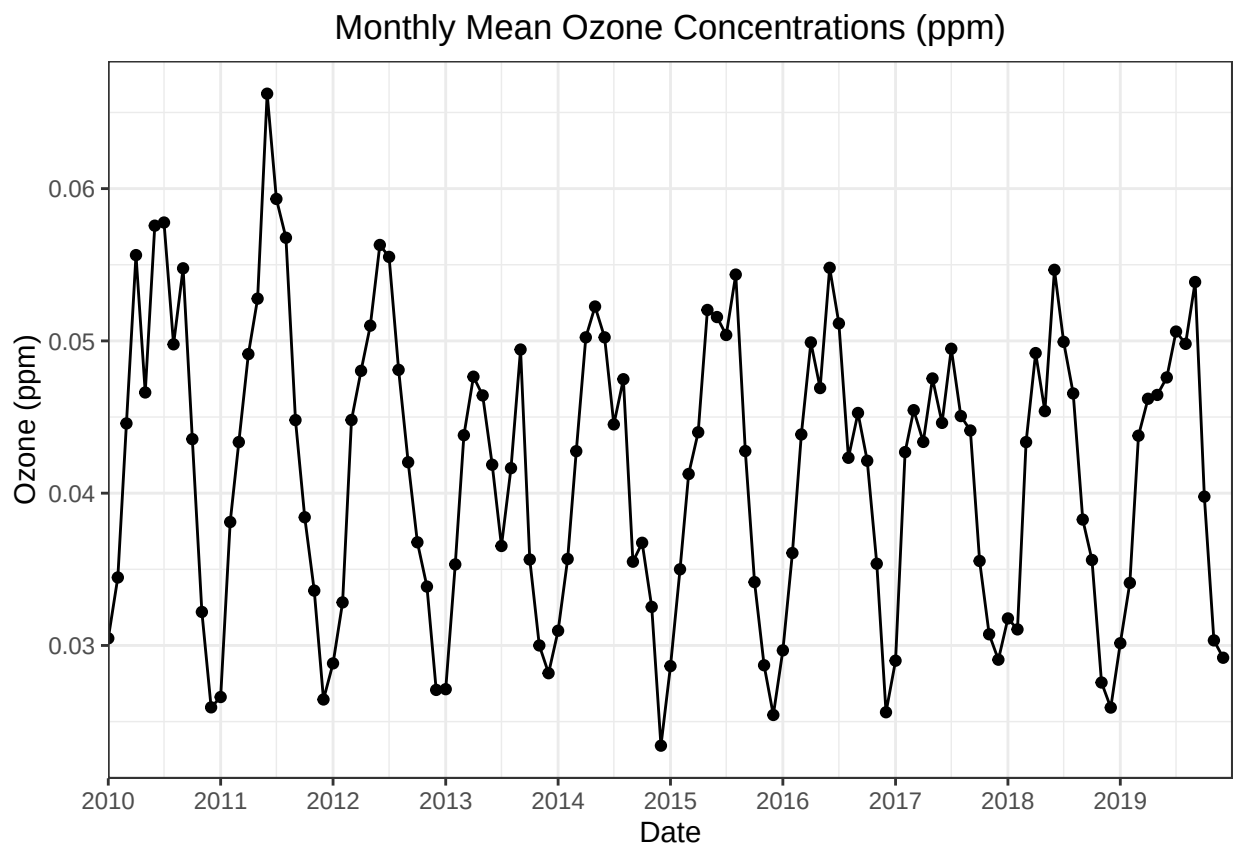
```
smk.test(GaringerOzone.monthly.ts)
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##      S varS
## -77 1499
```

Answer: It is because ozone concentrations exhibit strong seasonal patterns (higher in summer and lower in winter), and the test accounts for these recurring seasonal effects when assessing the long-term monotonic trend. This prevents seasonal cycles from being mistaken for an overall upward or downward trend.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly, aes(x = Date, y= meanO3))+
  geom_line() +
  geom_point()+
  labs(title = "Monthly Mean Ozone Concentrations (ppm)",
       x = "Date", y = "Ozone (ppm)")+
  scale_x_date(
    limits = c(as.Date("2010-01-01"),as.Date("2019-12-31")),
    expand = c(0,0),
    date_labels = "%Y",
    date_breaks = "1 year",
    date_minor_breaks = "6 month"
  )
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

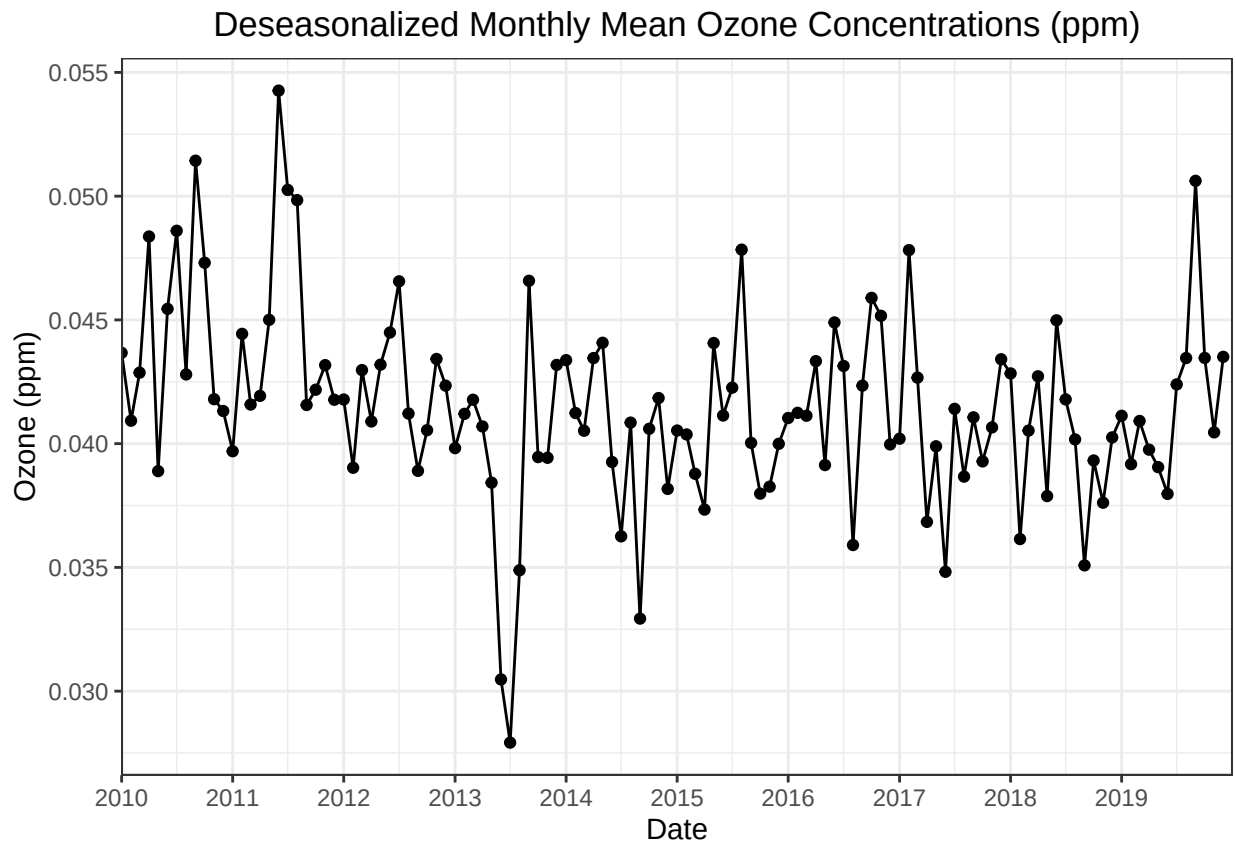
Answer: Ozone concentrations at the Garinger show strong seasonal variability, with peaks typically occurring in the warmer months and lower values in the cooler months. Accounting for this seasonal pattern, the seasonal Mann-Kendall trend test was performed and the result suggests that ozone levels at this site have decreased slightly but consistently throughout the decade ($z = -1.96$, $p = 0.0496$).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.

16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
GaringerOzone.monthly = GaringerOzone.monthly %>%
  ungroup() %>%
  mutate(mean03.nonseasonal = mean03 - monthly_decomp$time.series[,1])
ggplot(GaringerOzone.monthly, aes(x = Date, y = mean03.nonseasonal)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Deseasonalized Monthly Mean Ozone Concentrations (ppm)",
    x = "Date",
    y = "Ozone (ppm)"
  ) +
  scale_x_date(
    limits = c(as.Date("2010-01-01"), as.Date("2019-12-31")),
    expand = c(0, 0),
    date_labels = "%Y",
    date_breaks = "1 year",
    date_minor_breaks = "6 months"
  )
)
```

```
## Don't know how to automatically pick scale for object of type <ts>. Defaulting
## to continuous.
```



#16

```
GaringerOzone.monthly.nonseasonal.ts = ts(GaringerOzone.monthly$meanO3.nonseasonal,  
                                           start = c(2010,1), frequency = 12 )
```

```
mk.test(GaringerOzone.monthly.nonseasonal.ts)
```

```
##  
## Mann-Kendall trend test  
##  
## data: GaringerOzone.monthly.nonseasonal.ts  
## z = -2.9555, n = 120, p-value = 0.003121  
## alternative hypothesis: true S is not equal to 0  
## sample estimates:  
##          S          varS          tau  
## -1.304000e+03  1.943667e+05 -1.826331e-01
```

Answer: The nonseasonal Mann-Kendall test also detected a statistically significant downward trend in ozone concentrations over the 2010s ($z = -2.96$, $p = 0.003$). This result strengthens the finding from the Seasonal Mann-Kendall test on the complete (seasonal) series ($z = -1.96$, $p = 0.049$). By removing the recurring seasonal pattern, the trend becomes clearer and more significant, indicating that the overall decline in ozone concentrations is genuine and not driven by seasonal variation.